Trainings

Preparatory Steps

⚠ WARNING ⚠

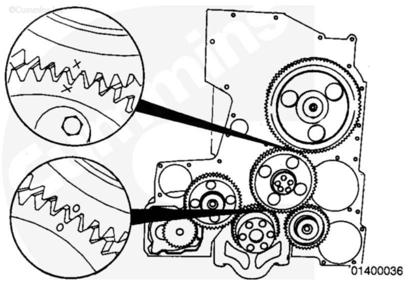
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

Remove

Note : The camshaft idler gear is the only idler gear that has timing marks.

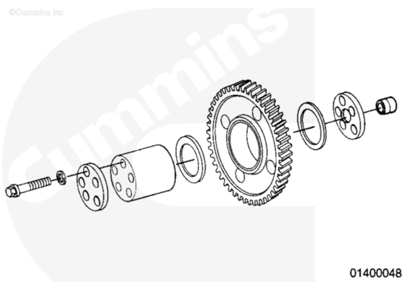
Align the "O" on the idler gear with the "O" on the crankshaft gear. Align the "X" on the camshaft gear with the "X" on the idler gear.



Remove the five capscrews, the retaining ring, and the front thrust bearing.

Remove the camshaft idler gear.

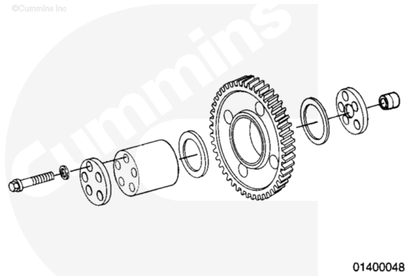
Remove the rear thrust bearing, the idler shaft, and the thrust bearing wear sleeve.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

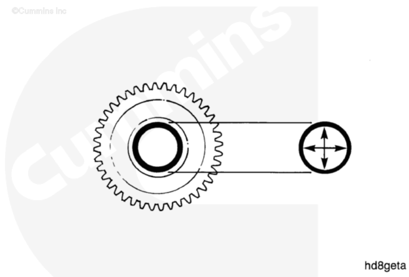
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent. Clean the idler gear and parts.

Dry with compressed air.

Check parts for damage.

Measure the inside diameter of the idler.



Idler Gear Inside Diameter		
mm		in
85.7758	MIN	3.3770
85.8012	MAX	3.3780

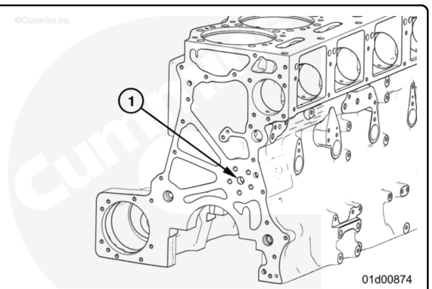
Check parts for damage.

Measure the outside diameter of the idler.

Idler Gear Bushing Outside Diameter		
mm		in
85.7123	MIN	3.3745
85.7250	MAX	3.3750

Note : The bushing in the gear is bored after installation. If machining capability is not available, replace the bushing and the gear as an assembly.

Measure the camshaft idler block bore diameter



Camshaft Idler Block Bore Diameter (1)		
mm		in
19.245	MIN	0.757
19.265	MAX	0.758

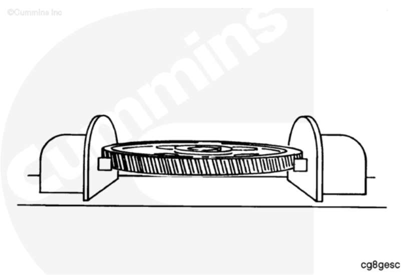
Magnetic Crack Inspect

⚠ **CAUTION** ⚠

Use copper braid contact that has neoprene covers to avoid burning the teeth of the gear.

⚠ **CAUTION** ⚠

If the gear contains a keyway, position the gear so that the keyway points toward one of the contacts when checking.



Use a magnetic particle testing machines.

Use the residual method. Apply head shot amperage.

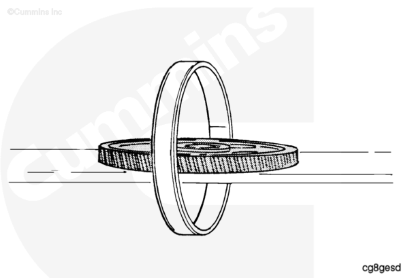
Adjust the amperage to the specified value

Gear Outside Diameter	Amperage D.C.
Less than 101 mm [4 inch]	1000
101 mm to 203 mm [4 inch to 8 inch]	1500
Greater than 203 mm [8 inch]	2000

Check the gear for cracks.

⚠ **CAUTION** ⚠

If the gear contains a keyway, position the gear so that the keyway points towards the coil.



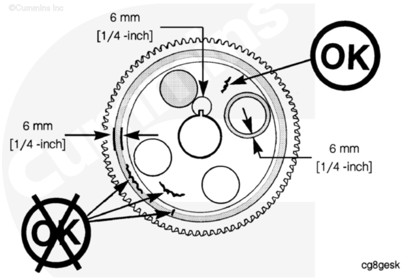
Use the residual methods. Apply coil shot amperage.

Adjust the amperage to the specified value.

Gear Outside Diameter	Amperage D.C.
Less than 101 mm [4 inch]	4000
101 mm to 203 mm [4 inch to 8 inch]	6000
Greater than 203 mm [8 inch]	8000

Note : Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check the gear for cracks.



Limits of acceptance - on Machined Surfaces

An open indication is visible to the eye after the wetting operation has been completed. An indication below the surface is **not** visible to the eye after the wetting operation has been completed. An indication below the surface can be seen with the use of ultraviolet light that is part of the machine.

Do **not** use the gear if:

- There is an open indication.
- There is an indication below the surface that is in the shaded area as shown.
- There is an indication below the surface that is longer than 6 mm [1/4 inch

Limits of acceptance - on Forged Surfaces

Do **not** use the gear if:

- There is an open indication that is in a circumferential direction.
- There is an open indication that is longer than 9.5 mm [3/8 inch].
- There is an indication below the surface that is in the shaded area.

An indication below the surface in a radial direction is acceptable (OK) in the area that is **not** shaded.

⚠ CAUTION ⚠

The gear must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

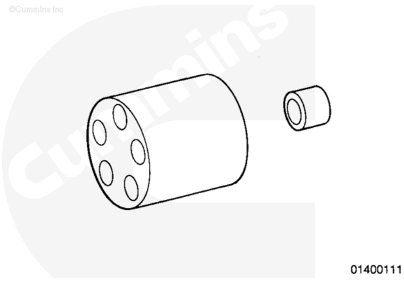
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

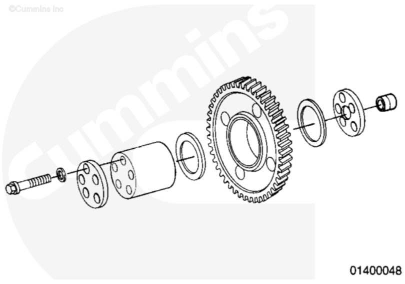
Demagnetize the gear

Use solvent or steam. Clean the part

Install



Note : If the idler shaft ring dowel remained in the cylinder block during disassembly, remove the dowel from the block and install it into the idler shaft. The bore in the thrust bearing wear plate and the outside diameter of the dowel are designed to provide a slight press fit to hold the wear plate in position during installation of the idler assembly.



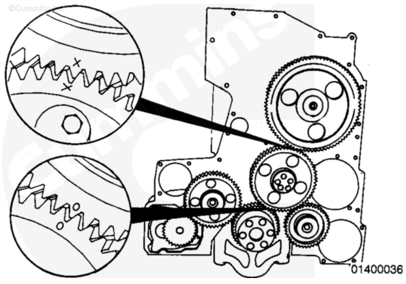
⚠ CAUTION ⚠

The grooves in the thrust washers must be positioned toward the gear.

Note : The timing marks on the camshaft gear must be visible when the gear is installed.

Use Lubriplate® Number 105, or equivalent. Lubricate the gear bushing, shaft, and the thrust washer.

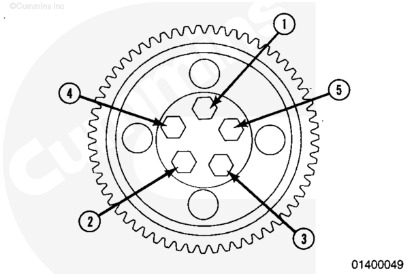
Use engine oil. Lubricate the capscrews. Assemble the parts as illustrated.



Note : The camshaft idler gear is the only idler gear that has timing marks.

Align the "O" on the idler gear with the "O" on the crankshaft gear. Align the "X" on the camshaft gear with the "X" on the idler gear.

Rotate the idler shaft as necessary to align the mounting capscrew holes.



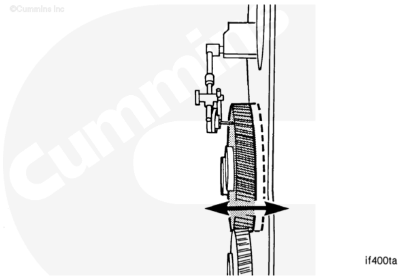
Use the capscrew to pull the shaft into the bore.

Use the following steps and tighten the capscrews in the sequence shown.

Torque Value:

1. 45 n•m [33 ft-lb]
2. 95 n•m [70 ft-lb]
3. 150 n•m [111 ft-lb]

Use a dial indicator. Measure the idler gear end clearance.



Camshaft Idler Gear - End Clearance		
mm		in
0.29	MIN	0.012
0.51	MAX	0.020

Note : If the clearance is not within specifications, check for foreign material between the parts, or check for proper location of the thrust washers. Oversize thrust washers are available.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the front gear cover. Refer to Procedure 001-031 in Section (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)
- Operate the engine. Check for leaks.